

$\Gamma \vdash \alpha \Rightarrow \beta$

The L.H. $\Rightarrow$ $\Gamma \vdash \alpha \Rightarrow \beta$ $\Rightarrow$ $\Gamma \vdash \alpha \Rightarrow \beta$

$\Gamma \vdash \alpha \Rightarrow \beta$ $\Rightarrow$ $\Gamma \vdash \alpha \Rightarrow \beta$ $\Rightarrow$ $\Gamma \vdash \alpha \Rightarrow \beta$

$$h \neq \alpha \rightarrow \neg \beta \rightarrow \neg (\phi \rightarrow \psi) \Leftrightarrow \neg \beta \rightarrow \neg (\phi \rightarrow \psi)$$

$$h(\alpha \rightarrow \neg \beta) \rightarrow \neg (\phi \rightarrow \psi) \Leftrightarrow \neg (\phi \rightarrow \psi)$$

$$h(\alpha \rightarrow \neg \beta) \rightarrow \neg (\phi \rightarrow \psi) \Leftrightarrow \neg (\phi \rightarrow \psi)$$

Handwritten mathematical diagram showing relationships between various expressions involving $\ln$ and α . The diagram includes terms like $\ln(\alpha)$, $\ln(p)$, $\ln(q)$, $\ln(B)$, and $\ln(p)$ with various arrows and underlines. Some terms are circled or crossed out.

Handwritten mathematical derivations for the distributive law of intersection over union. The image shows two main parts: a purple derivation at the top and a blue derivation at the bottom. Both parts start with the expression $(A \cap B) \cup (A \cap C)$ and aim to prove it is equal to $A \cap (B \cup C)$. The purple part uses a series of logical steps with set notation and logical symbols like $\wedge$, $\vee$, and $\subseteq$. The blue part uses a similar approach but includes a step where $(A \cap B) \cup (A \cap C)$ is shown to be a subset of $A \cap (B \cup C)$. The final result $A \cap (B \cup C)$ is highlighted in yellow in both parts.

Case 1: $\alpha \in \mathcal{L}$. $\neg(\alpha \wedge \neg\alpha)$

$$\Rightarrow \neg(\alpha \wedge \neg\alpha) \rightarrow (\neg\alpha \rightarrow \alpha) \quad \text{tautology}$$

$$\equiv 0 \rightarrow (\neg\alpha \rightarrow \neg\alpha) = 1$$

$$\equiv 1 \vee (\neg\alpha \rightarrow \neg\alpha) = 1$$

Case 2: $\alpha \in \mathcal{L}$. $\neg(\alpha \wedge \neg\alpha)$

$$\Rightarrow \neg(\alpha \wedge \neg\alpha) \rightarrow (\neg\alpha \rightarrow \alpha) \quad \text{tautology}$$

$$\Rightarrow \neg(\alpha \wedge \neg\alpha) \rightarrow (\neg\alpha \rightarrow \alpha) = 1$$

func:

$$\begin{aligned}
 & \left(\begin{array}{c} * \\ * \\ * \end{array} \right) \Leftrightarrow \tilde{\ln}(\alpha) \leq \ln(\beta) \rightarrow \\
 & \rightarrow (\ln \varphi) \leftrightarrow \ln(\beta) \Leftrightarrow \ln(\beta) = 1
 \end{aligned}$$

$$\Leftrightarrow \ln(\beta) \rightarrow (\ln \varphi \leftrightarrow 1) = 1$$

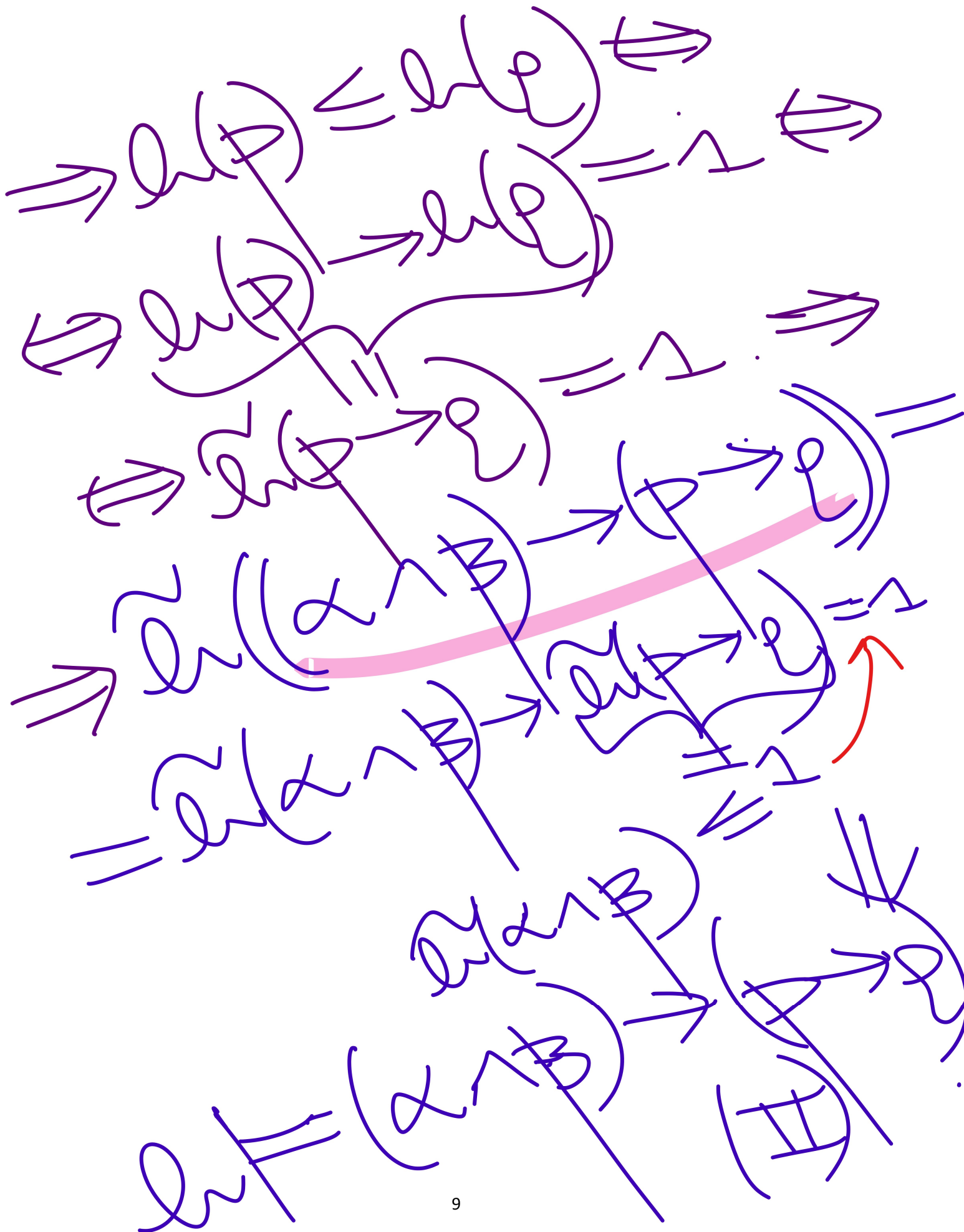
$$\Leftrightarrow \ln(\beta) \leq \ln(\varphi) \leftrightarrow 1 = 1$$

$$= (\ln \varphi) \rightarrow 1 \wedge$$

$$\wedge (1 \rightarrow \ln \varphi) =$$

$$= (\ln \varphi \vee 1) \wedge (1 \vee \ln \varphi) = 1$$

$$= 1 \wedge \ln(\varphi) = \ln(\varphi) \Rightarrow$$



$$(\text{I}), (\text{II}) \Rightarrow (\neg p: V \rightarrow \mathcal{L}_2)$$

$$(\neg (A \wedge B)) \rightarrow (\neg (A \rightarrow B))$$

$$\begin{aligned} & \Rightarrow \neg (A \wedge B) \rightarrow (\neg (A \rightarrow B)) \\ & \neg (A \wedge B) \rightarrow (\neg (A \rightarrow B)) \end{aligned}$$